

TABLE 2
POWER CONSUMPTION OF ATTINY13A + SSD1306 OLED

Mode/Condition	ATtiny13A Current	SSD1306 OLED Current	Total System Current (Typical)	Notes/Conditions
Power-down (deep sleep)	0.1-0.3 μ A	<10 μ A	~10 μ A	Watchdog disabled; OLED in sleep mode
Idle mode	~24 μ A (1MHz)	1-2mA (black screen)	~1-2mA	1.8V/1MHz; OLED ON but no pixels lit
Active mode (low frequency)	~190 μ A	8-12mA (typical content)	~8-12mA	1.8V/1MHz; typical text/icons displayed
Active mode (prototype design)	2.8-3.5mA	8-12mA (typical content)	6-7mA (measured)	2.7V/9.6MHz; moderate OLED usage
Active mode - Maximum OLED load	2.8-3.5mA	20-30mA (full white screen)	23-33mA	Worst case: all pixels ON, high contrast
Optimised low-power mode (intermittent)	~0.5-1mA (average)	~1-2mA (average)	~1-3mA (average)	Duty cycling; OLED updated every 10-60 seconds

TABLE 3
CHARGING TIME COMPARISON OF 10F AND 20F SUPERCAPACITORS

Harvesting Source/Scenario	Realistic Average Current	Time for 10F Supercapacitor	Time for 20F Supercapacitor	Typical Conditions/Notes
Solar – Full direct sunlight	40-50mA	8-10 minutes	16-20 minutes	Peak output from 3V, 50mA solar cell under strong sunlight
Solar – Bright outdoor/indirect sunlight	20-30mA	13-20 minutes	26-40 minutes	Cloudy or shaded outdoor conditions
Solar – Good indoor/office light	5-10mA	33-80 minutes	1.1-2.7 hours	500-1000 lux ambient indoor lighting
Solar – Low/dim indoor light	1-3mA	2.8-8 hours	5.6-16 hours	Poor lighting with a small solar cell
Vibration/piezo (strong vibration)	100-500 μ A	2.8-14 hours	5.6-28 hours	Continuous mechanical vibration (machines, walking)
Vibration/piezo (moderate/occasional)	10-50 μ A	1.1-11 days	2.2-22 days	Intermittent human motion or light machinery
Electromagnetic/coil (strong field)	50-200 μ A	2.8-11 days	5.6-22 days	Near high-current AC lines, motors, or transformers
Electromagnetic/coil (weak field)	5-20 μ A	14-56 days	28-112 days	Ambient electromagnetic noise from distant sources
RF/Ambient radio waves	1-10 μ A	28-140 days	56-280 days	Near Wi-Fi routers or cell towers (very low harvested power)

EFY Note: These charging times are based on theoretical calculations and ideal assumptions. Actual charging times may vary significantly under real-world conditions due to fluctuations in energy availability, conversion efficiency, and system losses.

design. One IC was optimised for harvesting energy from direct sunlight and indoor ambient light, while the other was dedicated to electromagnetic induction and vibration-based energy harvesting.

Energy harvested from both sources was combined and stored in the supercapacitor, which continuously powered the microcontroller. This enhanced approach enabled approximately 20-25 minutes of uninterrupted operation without requiring a battery. The components required to build this system are listed in Table 1.

In many modern batteryless devices, numerous ultra-low-power MCUs are available. Some are BLE-based and consume extremely low power, often in the microamp range or even nanoamp range in deep-sleep modes, while operating at 3V or 1.8V supply voltages. For simplicity and easy availability,

the ATtiny13A microcontroller was selected for this design.

According to the datasheet, the device offers very low power consumption across different operating modes. In active mode, it typically consumes around 190 μ A at 1.8V and 1MHz. In idle mode under the same conditions, the current drops to approximately 24 μ A. In power-down mode, with the watchdog timer disabled, current consumption is typically less than 0.1-0.3 μ A. These low-power characteristics make the ATtiny13A suitable for energy-harvesting and batteryless applications.

During prototype testing, the ATtiny13A operated at its maximum internal clock frequency of 9.6MHz with a supply voltage of 2.7V provided by the charged supercapacitor. At these higher frequencies and voltages, current consumption increased significantly, as power scales approximately linearly with

clock frequency and supply voltage. The measured active-mode current ranged from 2.8-3.5mA.

The SSD1306 OLED display contributes a significant portion of the total power consumption. In sleep mode, current is below 10 μ A. With a completely black screen, it draws 1-2mA. Typical text or icon display (25-50% of pixels lit) consumes 8-12mA, while a full white screen can increase consumption to 20-30mA.

Testing showed that the combined ATtiny13A-SSD1306 system consumed 6-7mA continuously during moderate display activity. This allowed the 2.7V, 3F supercapacitor to power the system for up to five minutes in normal operation and up to 20-25 minutes in optimised low-power modes using reduced update frequency, lower contrast, or brief sleep intervals. Table 2 summarises the